

Nobel Prize-winning neuroscientist Santiago Ramón y Cajal used microscopes to explore the organization of the nervous system. He was the first to realize that it is composed of independent cells that communicate with one another.

MILESTONES

TURNING POINT

Shown Golgi's obscure tissue-staining method in 1887 and realizes it can further his research.

TECHNOLOGICAL FEAT

Modifies Golgi's stain in 1888, and uses it to illuminate human brain tissue.

DISCREDITS THEORY

In 1888, shows nerve cells are not joined in continuous web, as in reigning theory, but are separate units.

WRITES MAGNUM OPUS

Publishes the first volume of his major book on the structure of the nervous system in 1899.

PRIZE WINNER

With Golgi, jointly wins Nobel prize in 1906 for his work on the structure of the nervous system.

Born in northern Spain, Santiago Ramón y Cajal was a rebellious child with a passion for drawing. He wanted to be an artist, but his father, a practicing doctor who also taught anatomy, persuaded him to go to medical school. Eventually, Cajal's artistic talents and scientific interests merged perfectly in the fast-developing field of neuroscience.

Microscopes and staining

While studying to become a doctor, Cajal used a microscope for the first time. Impressed, he used his savings to buy one and began to view and draw the structure of the muscle tissue that he saw on an increasingly small scale. In 1885, he was given a more modern microscope, and his work became entirely focused on histology—anatomy at microscopic level. Before long, he had started to investigate the nervous system.

Prior to the 1870s, using a microscope to examine nervous tissue had had limitations—low magnification and poor resolution made it hard to determine where one structure ended and another began. But this changed with advances in optical lenses and better versions of the stains that were applied to tissue samples in order to make individual cells more visible.

In 1887, while professor of histology and pathological anatomy at the University of Barcelona, Cajal saw brain tissues that had been stained using

Having modified Golgi's tissue-staining technique, Cajal produced hundreds of illustrations of the human brain and nervous system.

